§2 Exercises

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2.1. Vector fields
Let X be the vector field $x\partial/\partial x + y\partial/\partial y$ and $f(x,y,z)$ the function $x^2+y^2+z^2$ on $\mathbb{R}^3$. Compute Xf.

Proof. By definition, given $p = (p^1, p^2, p^3) \in A$,

$$\begin{aligned}(Xf)(p) &= X_p f \\ &= p^1 \frac{\partial f}{\partial x}\bigg|_p + p^2 \frac{\partial f}{\partial y}\bigg|_p \\ &= p^1 * (2x)\bigg|_p + p^2 * (2y)\bigg|_p \\ &= 2(p^1)^2 + 2(p^2)^2.\end{aligned}$$

Thus, in general, $(Xf)(x,y,z) = 2x^2 + 2y^2$. □

2.2. Algebra structure on C_p^∞
Define carefully addition, multiplication, and scalar multiplication in C_p^∞. Prove that addition in C_p^∞ is commutative.

Proof. Given $[f], [g]$ germs in C_p^∞, where f and g are representatives of $[f]$ and $[g]$ respectively, define $[f] + [g] = [f+g]$. To show that this is well-defined, consider $f_1 \in [f]$ and $g_1 \in [g]$. Let U (resp. V) be a neighborhood on which f and f_1 (resp. g and g_1) agree, and note that $U \cap V$ is a neighborhood of p. Then for some $q \in U \cap V$,

$$\begin{aligned}(f+g)(q) &= f(q) + g(q) \\ &= f_1(q) + g_1(q) \\ &= (f_1 + g_1)(q).\end{aligned}$$

Thus $(f+g)|_{U\cap V} = (f_1+g_1)|_{U\cap V}$, so $[f+g] = [f_1+g_1]$. From this, it is immediate that addition is commutative as defined, since

$$[f] + [g] = [f+g] = [g+f] = [g] + [f].$$

Similarly, define $[f] * [g] = [f * g]$. The proof that this is well-defined is the same as above, swapping out each "$+$" for a "$*$".

Finally, for $a \in \mathbb{R}$, $[f] \in C_p^\infty$, and some x in the domain of f, define $(af)(x) = a * f(x)$. □

2.3. Vector space structure on derivations at a point
Let D and D' be derivations at p in $\mathbb{R}^n$, and $c \in \mathbb{R}$. Prove that

(a) the sum $D + D'$ is a derivation at p.

(b) the scalar multiple cD is a derivation at p.

Proof. (a) Given $[f], [g] \in C_p^\infty$, $f \in [f]$, $g \in [g]$, and $c \in \mathbb{R}$,

$$\begin{aligned}(D + D')([f] + [g]) &= D([f] + [g]) + D'([f] + [g]) \\ &= D([f]) + D([g]) + D'([f]) + D'([g]) \\ &= D([f]) + D'([f]) + D([g]) + D'([g]) \\ &= (D + D')([f]) + (D + D')([g]),\end{aligned}$$

so $D + D'$ is additive. Also,

$$\begin{aligned}(D + D')(c[f]) &= D(c[f]) + D'(c[f]) \\ &= cD([f]) + cD'([f]) \\ &= c\,[D([f]) + D'([f])] \\ &= c(D + D')([f]),\end{aligned}$$

so $D + D'$ is homogeneous. Finally,

$$\begin{aligned}(D + D')([f][g]) &= D([f][g]) + D'([f][g]) \\ &= D([f])g(p) + f(p)D([g]) + D'([f])g(p) + f(p)D'([g]) \\ &= g(p)\,[D([f]) + D'([f])] + f(p)\,[D([g]) + D'([g])] \\ &= g(p)(D + D')([f]) + f(p)(D + D')([g]),\end{aligned}$$

so $D + D'$ follows the Leibniz rule, and thus is a derivation at p.

(b) Given $[f], [g]$ as above and $c, d \in \mathbb{R}$,

$$\begin{aligned}(cD)([f] + [g]) &= c(D([f] + [g])) \\ &= c(D([f]) + D([g])) \\ &= cD([f]) + cD([g]) \\ &= (cD)([f]) + (cD)([g]),\end{aligned}$$

$$\begin{aligned}(cD)(d[f]) &= c(D(d[f])) \\ &= c(dD([f])) \\ &= d(cD)([f]),\end{aligned}$$

and

$$\begin{aligned}(cD)(ab) &= c(D(ab)) \\ &= c(D(a)b + aD(b)) \\ &= cD(a)b + caD(b) \\ &= (cD)(a)b + a(cD)(b).\end{aligned}$$

$\square$

2.4. Product of derivations
Let A be an algebra over a field K. If D_1 and D_2 are derivations of A, show that $D_1 \circ D_2$ is not necessarily a derivation (it is if D_1 or $D_2 = 0$), but $D_1 \circ D_2 - D_2 \circ D_1$ is always a derivation of A.

Proof. Given $a, b \in A$,

$$\begin{aligned}(D_1 \circ D_2)(ab) &= D_1(D_2(ab)) \\ &= D_1(D_2(a)b + aD_2(b)) \\ &= D_1(D_2(a)b) + D_1(aD_2(b)) \\ &= D_1(D_2(a))b + D_2(a)D_1(b) + D_1(a)D_2(b) + aD_1(D_2(b)), \quad (1)\end{aligned}$$

so $D_1 \circ D_2$ can only satisfy the Leibniz rule if $D_2(a)D_1(b) + D_1(a)D_2(b) = 0$. If D_1 or $D_2 = 0$, certainly this holds, but to see that it does not hold in general, consider $A = C^\infty(\mathbb{R})$ and $D_1 = D_2 = d/dx$. Then taking $a = x$ and $b = x^2$ where each is a function of x from $\mathbb{R}$ to $\mathbb{R}$,

$$D_2(a)D_1(b) + D_1(a)D_2(b) = (1)(2x) + (1)(2x) = 4x,$$

which is nonzero.

Now we use (1) from above to show that the Leibniz rule holds for $D_1 \circ D_2 - D_2 \circ D_1$:

$$\begin{aligned}&(D_1 \circ D_2 - D_2 \circ D_1)(ab) \\ &= D_1(D_2(a))b + D_2(a)D_1(b) + D_1(a)D_2(b) + aD_1(D_2(b)) \\ &\quad - [D_2(D_1(a))b + D_1(a)D_2(b) + D_2(a)D_1(b) + aD_2(D_1(b))] \\ &= (D_1 \circ D_2)(a) * b + a(D_1 \circ D_2)(b) - (D_2 \circ D_1)(a) * b - a(D_2 \circ D_1)(b) \\ &= (D_1 \circ D_2 - D_2 \circ D_1)(a) * b + a(D_1 \circ D_2 - D_2 \circ D_1)(b),\end{aligned}$$

demonstrating that the problematic "middle" terms from before cancel out, and we are left with a clean statement of the Leibniz rule. ☐